arXiv:2604.17335v2 [cs.RO] 12 Jul 2026

Learning Whole-Body Humanoid Locomotion via Motion Generation and Motion Tracking

Zewei Zhang[1,4], Kehan Wen[1], Michael Xu[2], Junzhe He[1], Chenhao Li[1,3], Takahiro Miki[1], Clemens Schwarke[1,5], Chong Zhang[1,3], Xue Bin Peng[2,5], and Marco Hutter[1]

***Abstract*—Whole-body humanoid locomotion is challenging due to high-dimensional control, morphological instability, and the need for real-time adaptation to various terrains using onboard perception. Directly applying reinforcement learning (RL) with reward shaping to humanoid locomotion often leads to lower-body-dominated behaviors, whereas imitation-based RL can learn more coordinated whole-body skills but is typically limited to replaying reference motions without a mechanism to adapt them online from perception for terrain-aware locomotion. To address this gap, we propose a whole-body humanoid locomotion framework that combines skills learned from reference motions with terrain-aware adaptation. We first train a diffusion model on retargeted human motions for real-time prediction of terrain-aware reference motions. Concurrently, we train a whole-body reference tracker with RL using this motion data. To improve robustness under imperfectly generated references, we further fine-tune the tracker with a frozen motion generator in a closed-loop setting. The resulting system supports directional goal-reaching control with terrain-aware whole-body adaptation, and can be deployed on a Unitree G1 humanoid robot with onboard perception and computation. The hardware experiments demonstrate successful traversal over boxes, hurdles, stairs, and mixed terrain combinations. Quantitative results further show the benefits of incorporating online motion generation and fine-tuning the motion tracker for improved generalization and robustness.**

I. Introduction

Developing whole-body perceptive humanoid locomotion remains a challenging yet important step toward expanding the operational range of humanoid robots. While deep reinforcement learning (RL) has achieved remarkable success in enabling highly dynamic behaviors on quadrupedal systems [1–6], extending these capabilities to humanoid robots remains considerably difficult, as humanoids possess higher degrees of freedom together with more complex morphological constraints. As a result, training an RL-based whole-body humanoid controller solely through reward shaping can be non-trivial and often suffers from inefficient exploration. Without explicit structural guidance, vanilla RL-based humanoid controllers frequently converge to locomotion strategies with limited whole-body coordination, making it difficult to acquire coordinated behaviors for challenging terrain traversal. This limitation becomes especially evident in tasks such as humanoid parkour, where coordinated whole-body movements are essential for traversing large and complex obstacles.

To alleviate the difficulty of reward shaping and inefficient exploration, motion imitation via RL has emerged as a promising alternative [7–12]. By leveraging reference data, motion tracking can efficiently transfer highly coordinated whole-body skills to the robot. However, pure motion trackers are inherently limited to replaying choreographed trajectories and generally lack the adaptability and reactivity required for operation in unstructured and diverse environments. Achieving human-level perceptive locomotion therefore requires a higher-level composition mechanism that can dynamically adjust locomotion style across varied obstacles based on real-time perceptive inputs.

One direct solution for skill composition is to distill multiple specialized teacher policies [13], or to compose multiple skills from separate reference trajectories within a single imitation-based teacher framework [14, 15]. By learning distinct expert skills for different terrain conditions, a unified distilled policy can learn to transition among them using exteroceptive observations. However, current methods have not yet demonstrated strong scalability to a diverse set of skills or robust generalization to new terrains.

Alternatively, generative models have recently shown strong potential for scalable motion control. Training generative models on large-scale motion datasets has proven effective in both character control [16–18] and robot control [9, 19], naturally reducing the engineering effort required as the motion repertoire expands. However, directly deploying expressive generated kinematic trajectories on real humanoid locomotion systems may introduce motion artifacts, such as foot sliding or temporal discontinuities, which can lead to catastrophic failure. To mitigate such issues, recent work [18, 20, 21] combines diffusion-based motion generation with RL-based motion tracking, allowing the tracker to output physically plausible actions based on kinematic generation. Despite this progress, their applicability to real humanoid robots in whole-body locomotion tasks remains insufficiently explored, and their deployment may potentially be hindered by slow inference and limited robustness to real-world disturbances.

To address these limitations, we propose a whole-body humanoid locomotion system that adapts a motion generation and motion tracking framework to real-time robot control for challenging humanoid locomotion tasks. In particular, we repurpose coordinated human whole-body skills learned from motion data for perceptive humanoid locomotion, and build a control framework that combines diffusion-based motion generation with RL-based motion tracking. The resulting system supports

Manuscript received: April 9, 2026; Accepted: June 18, 2026. This paper was recommended for publication by Editor Olivier Stasse upon evaluation of the Associate Editor and Reviewers comments.

[1]Robotic Systems Lab, ETH Zurich {zewzhang, kehwen, junzhe, chenhli, tamiki, cschwarke, chozhang, mahutter}@ethz.ch
[2]Simon Fraser University {mxa23, xbpeng}@sfu.ca
[3]ETH AI Center. [4]EPFL. [5]NVIDIA.

Digital Object Identifier (DOI): see top of this page.

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通过运动生成与运动跟踪学习全身人形机器人运动

Zewei Zhang[1,4], 、Kehan Wen[1], 、Michael Xu[2]、Junzhe He[1]、Chenhao Li[1,3]、Takahiro Miki[1]、Clemens Schwarke[1,5], 、Chong Zhang[1,3]、Xue Bin Peng[2,5],和 Marco Hutter[1]

摘要—全身人形机器人运动具有挑战性，原因在于高维控制、形态不稳定性，以及需要利用机载感知对不同地形进行实时适应。将带有奖励塑形的强化学习（***RL***）直接应用于人形机器人运动，往往会导致以腿部为主导的行为；而基于模仿的***RL***虽然能够学习更协调的全身技能，但通常只能回放参考运动，缺乏从感知中在线调整参考运动以实现地形感知运动的机制。针对这一不足，我们提出了一种全身人形机器人运动框架，将参考运动中学到的技能与地形感知自适应结合起来。我们首先在重定向后的人体运动上训练扩散模型，以实时预测地形感知的参考运动。同时，我们利用这些运动数据，通过***RL***训练全身参考运动跟踪器。为了在参考生成不完善的情况下提升鲁棒性，我们进一步在闭环设置中冻结运动生成器，对跟踪器进行微调。最终系统支持带地形感知全身自适应的定向目标到达控制，并可部署在配备机载感知与计算的宇树***G1***人形机器人上。硬件实验表明，该系统能够成功通过箱子、栏架、楼梯以及混合地形组合。定量结果进一步表明，引入在线运动生成并对运动跟踪器进行微调，有助于提高泛化能力和鲁棒性。

I. 引言

开发全身感知的人形机器人运动，仍然是拓展人形机器人作业范围的重要且具挑战性的步骤。虽然深度强化学习（RL）已在四足系统 [1–6], 上实现高度动态行为方面取得显著成功，但将这些能力扩展到人形机器人上仍然相当困难，因为人形机器人具有更高的自由度以及更复杂的形态约束。因此，仅通过奖励塑形来训练基于RL的全身人形控制器并非易事，且常常面临探索效率低下的问题。在没有显式结构引导的情况下，普通基于RL的人形控制器往往收敛到全身协调性有限的运动策略，从而难以获得穿越挑战性地形所需的协调行为。这一局限性在人形跑酷等任务中尤为明显，在这些任务中，协调的全身运动对于穿越大型复杂障碍物至关重要。

为了缓解奖励塑形的困难以及探索效率低下的问题，基于强化学习（RL）的运动模仿已成为一种很有前景的替代方案 [7–12]。通过利用参考数据，运动跟踪能够高效地将高度协调的全身技能迁移到机器人上。然而，纯运动跟踪方法本质上仅限于重放预先编排好的轨迹，通常缺乏在非结构化和多样化环境中运行所需的适应性与响应性。因此，要实现人类水平的感知运动，需要一种更高层级的组合机制，从而能够根据实时感知输入，在各种障碍物环境中动态调整运动风格。

技能组合的一种直接解决方案是，将多个专门的教师策略 [13]，蒸馏到统一策略中，或者在一个统一的基于模仿的教师框架 [14, 15] 中，从不同的参考轨迹组合出多种技能。通过学习不同地形条件下的不同专家技能，一个统一的蒸馏策略可以利用外部感知观测在这些技能之间进行转换。然而，现有方法尚未展现出对多样化技能集的强大可扩展性，也未展现出对新地形的稳健泛化能力。

另外，生成模型近来在可扩展运动控制方面展现出强大的潜力。在大规模运动数据集上训练生成模型已被证明是有效的，在角色控制 [16–18] 和机器人控制 [9, 19], 中均如此，并且所需的工程投入会随运动技能库的扩展而自然减少。然而，将富有表现力的生成式运动学轨迹直接部署到真实人形机器人运动系统上，可能会引入足部滑动或时间不连续等运动伪影，从而导致灾难性失败。为缓解这些问题，近期工作 [18, 20, 21] 将基于扩散的运动生成与基于强化学习的运动跟踪相结合，使跟踪器能够基于运动学生成结果输出物理上合理的动作。尽管取得了上述进展，这些方法在真实人形机器人的全身运动任务中的适用性仍未被充分探索，而且其部署可能因推理速度慢、对现实世界扰动的鲁棒性有限而受到阻碍。

为了解决这些局限性，我们提出了一种全身人形机器人运动系统，该系统将运动生成和运动跟踪框架适配到实时机器人控制中，以应对具有挑战性的人形机器人运动任务。具体而言，我们将从运动数据中学习到的协调的人体全身技能重新用于感知型人形机器人运动，并构建了一个结合基于扩散的运动生成和基于强化学习的运动跟踪的控制框架。由此产生的系统支持

收稿日期：2026年4月9日；录用日期：2026年6月18日。本文由编辑 Olivier Stasse 在评估副编辑和审稿人的意见后推荐发表。[1]苏黎世联邦理工学院机器人系统实验室 {zewzhang, kehwen, junzhe, chenhli, tamiki, cschwarke, chozhang, mahutter}@ethz.ch[2]西蒙弗雷泽大学 {mxa23, xbpeng}@sfu.ca[3]苏黎世联邦理工学院人工智能中心。[4]洛桑联邦理工学院。[5]英伟达。数字对象标识符（DOI）：见本页顶部。



Fig. 1. Whole-body humanoid locomotion on the Unitree G1 robot enabled by diffusion-based motion generation and RL-based motion tracking. Our hardware experiments show behaviors including box climbing, box descent, vaulting, and traversal over mixed terrain sequences. https://wholebodylocomotion.github.io/.

directional goal-reaching control together with terrain-aware whole-body adaptation. Our hardware experiments show that our approach enables efficient skill execution and generalization across diverse terrains, including terrains unseen during training. Our results further demonstrate the effectiveness of using a motion generator trained on human motion data to guide an RL-based controller through challenging terrains requiring diverse skills, without heavily engineered distillation strategies.

II. Related work

A. Whole-body Motion Tracking for Humanoids

In simulated whole-body character control, prior works [7, 22] have explored RL-based frameworks in which characters learn to imitate reference motion clips. By training a control policy to minimize the discrepancy between the reference motion and the executed motion, the controller is able to reproduce a wide range of dynamic maneuvers while remaining consistent with the physical constraints of the controlled system.

More recently, this paradigm has been widely adopted in quadruped and humanoid robot control [9, 12, 23, 24]. By designing imitation rewards and regularization terms, and optimizing the policy with the PPO algorithm [25], recent works have demonstrated strong capability in imitating large-scale motion datasets and even enabling real-time teleoperation [11, 26, 27]. However, for general locomotion tasks, pure motion tracking remains fundamentally limited in its adaptability to varying terrains.

B. Motion Generation for Motion Control

Motion generation with generative models provides an effective way to sample feasible motion trajectories from curated motion datasets. Prior work [17] has demonstrated that diffusion models can generate smooth kinematic motion trajectories conditioned on text descriptions by incorporating additional geometric losses. Building on this line of work, Tevet et al. [18] further extend it to physical character control by coupling the generated motion with an RL-based motion tracker, which helps correct artifacts introduced by the generative model. Serifi et al. [20] also combine motion generation with RL-based motion tracking, and further fine-tune the generator using value information from the critic network of the tracking policy. Xu et al. [21] adopt a similar design philosophy to [18] and extend it to humanoid parkour in simulation, demonstrating impressive whole-body motion control conditioned on surrounding terrain observations. However, these methods mainly target simulated character control, and may still be constrained by the inference latency of diffusion models and lack of robustness for real robotic systems. As a result, relatively few real-time whole-body humanoid locomotion systems with integrated exteroceptive sensing have been demonstrated.

Another line of work [9, 16] generates joint state-action pairs without relying on a separate motion tracker. These methods have been demonstrated on humanoid robots for tasks such as waypoint navigation and motion inpainting. However, because control is produced directly from the co-diffusion process, they may still suffer from motion artifacts that can lead to unstable locomotion. Moreover, more dynamic and demanding tasks, such as humanoid whole-body locomotion, are not well-studied in this setting. In our work, we follow the general framework of [21], and adapt it to real-world humanoid locomotion tasks that require both whole-body coordination and real-time reaction to terrain conditions.

C. Skill Composition for Perceptive Locomotion Control

Composing multiple skills into a unified policy is often regarded as an effective shortcut toward whole-body visuomotor locomotion control, compared with training a single policy entirely from scratch. Hoeller et al. [28] first train multiple expert skills for quadrupedal locomotion across different terrain types, and then learn a high-level planner that switches categorically among the expert policies. In contrast, Rudin et al. [4] employ DAgger [29] to first distill several expert policies into a single policy, and subsequently fine-tune the distilled policy with RL on new terrains.

For humanoid whole-body locomotion, concurrent work [13] adopts a similar strategy to [4]: it first trains terrain-specific expert policies from scratch with RL and then distills them into a unified student policy. Wu et al. [14] likewise apply a similar distillation framework, but construct the expert skills from imitation-based RL policies trained with whole-body motion references. These methods have shown promising results for terrain traversal requiring coordinated whole-body control. However, both approaches may rely on carefully designed distillation procedures, such as expert assignment and data distribution design, to efficiently learn transitions and combinations across different skills.

In our work, instead of explicitly engineering a distillation pipeline over multiple expert skills, we use a diffusion-based



图 1 基于扩散的运动生成与基于强化学习的运动跟踪，使宇树 G1 人形机器人实现了全身运动。我们的硬件实验展示了爬箱、下箱、跨越以及穿越混合地形序列等行为。https://wholebodylocomotion.github.io/。

定向目标到达控制以及基于地形感知的全身自适应。我们的硬件实验表明，我们的方法能够实现高效的技能执行，并能在多样地形（包括训练期间未见过的地形）上实现泛化。我们的结果进一步证明了使用基于人类运动数据训练的运动生成器来指导基于RL的控制器穿越需要多种技能的挑战性地形的有效性，且无需大量精心设计的蒸馏策略。

II. 相关工作

A. 人形机器人的全身运动跟踪

在仿真的全身角色控制中，先前的工作 [7,22] 探索了基于强化学习的框架，其中角色学习模仿参考运动片段。通过训练控制策略来最小化参考运动与执行运动之间的差异，控制器能够在与被控系统的物理约束保持一致的同时，复现各种动态动作。

最近，这种范式已被广泛应用于四足机器人和人形机器人的控制 [9, 12, 23, 24]。通过设计模仿奖励和正则化项，并使用 PPO 算法 [25]，优化策略，近期的工作已经展示了在模仿大规模运动数据集方面的强大能力，甚至实现了实时遥操作 [11, 26, 27]。然而，对于一般的运动任务，纯运动跟踪在适应不同地形方面仍存在根本性局限。

B. 面向运动控制的运动生成

利用生成模型进行运动生成，为采样可行运动轨迹提供了一种有效方式，这些轨迹采样自精选运动数据集。先前工作 [17]已经证明，扩散模型通过加入额外的几何损失，能够生成以文本描述为条件的平滑运动学运动轨迹。在这一系列工作基础上，Tevet等人 [18] 通过将生成的运动与基于强化学习的运动跟踪器耦合，进一步将其扩展到物理角色控制，从而帮助修正生成模型引入的伪影。Serifi等人 [20] 同样将运动生成与基于强化学习的运动跟踪相结合，并利用跟踪策略的评论家网络提供的价值信息进一步微调生成器。Xu等人 [21]采用了与 [18]类似的设计理念，并将其扩展到仿真中的人形跑酷，展示了以周围地形观测为条件的令人印象深刻的全身运动控制。然而，这些方法主要面向仿真角色控制，可能仍受限于扩散模型的推理延迟，且对真实机器人系统缺乏鲁棒性。因此，目前已演示的、集成外部感知的实时全身人形运动系统相对较少。

另一类工作 [9, 16] 直接生成关节状态-动作对，而无需依赖独立的运动跟踪器。这些方法已在人形机器人上得到验证，可用于诸如航点导航和运动修复等任务。然而，由于控制输出直接来自联合扩散过程，它们仍可能产生运动伪影，进而导致运动不稳定。此外，更具动态性和挑战性的任务，例如人形机器人全身运动，在这种设定下尚未得到充分研究。在我们的工作中，我们遵循[21], 的总体框架，并将其适配到需要全身协调并实时响应地形条件的真实世界人形运动任务。

C. 面向感知运动控制的技能组合

与完全从头训练单一策略相比，将多个技能组合成一个统一策略通常被视为实现全身视觉引导运动控制的有效捷径。Hoeller等人 [28] 首先针对不同类型的地形训练多个四足运动专家技能，然后学习一个高层规划器，在专家策略之间进行离散切换。相比之下，Rudin等人 [4] 采用 DAgger [29] ，首先将多个专家策略蒸馏为单一策略，随后在新地形上使用强化学习微调蒸馏后的策略。

对于人形机器人的全身运动，同期工作 [13]采取了与 [4]相似的策略：首先使用强化学习从零开始训练特定地形的专家策略，然后将它们蒸馏为一个统一的学生策略。Wu等人 [14] 同样应用了类似的蒸馏框架，但专家技能来自基于模仿的强化学习策略，这些策略使用全身运动参考进行训练。这些方法在需要协调全身控制的地形穿越方面已展现出令人鼓舞的结果。然而，这两种方法可能都依赖于精心设计的蒸馏流程（如专家分配和数据分布设计），才能高效地学习不同技能之间的转换与组合。

在我们的工作中，我们没有显式地设计一个涵盖多种专家技能的蒸馏流程，而是使用一种基于扩散的

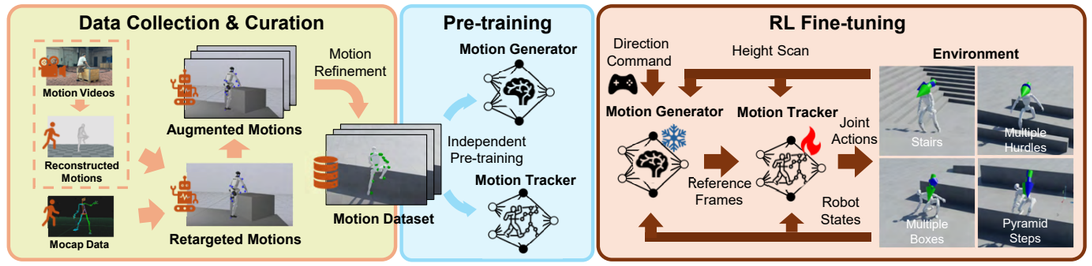


Fig. 2. Overview of the training framework. (a) Data Collection & Curation: whole-body robot motions are obtained from human motions through motion reconstruction and retargeting, followed by motion augmentation. (b) Pre-training: the diffusion-based motion generator and RL-based motion tracker are trained solely on the **offline motion dataset** from stage (a). (c) RL Fine-tuning: the motion tracker is **fine-tuned with RL** under randomized direction commands and on more diverse terrains with randomized configurations, including stairs, multiple vaulting hurdles, boxes, and pyramid-shaped high steps, while running the frozen motion generator.

motion generator trained on retargeted human motion data as a skill composition module that produces corresponding motion references for a low-level tracking policy based on the environment conditions. In addition, to reduce the impact of motion artifacts introduced by the generator and improve control performance, we further fine-tune the motion tracking policy with RL under randomized direction commands and more diverse terrain configurations while using a frozen motion generator pretrained on the offline motion dataset. More details are provided in Sec. III-C.

III. Method

The overall framework is illustrated in Fig. 2. Our method consists of three stages: (a) collecting whole-body motion data for different terrains and converting them into dynamically feasible robot trajectories; (b) pre-training the RL-based motion tracker and the diffusion-based motion generator on this dataset; and (c) fine-tuning the motion tracker on more diverse terrain configurations, while keeping the motion generator frozen and running it in a receding-horizon manner during deployment. We describe each stage below.

A. Data Collection & Curation

Our initial motion data consist of approximately 5 minutes of motion clips collected from two sources: motion videos captured by ourselves and public motion datasets [12, 30]. The dataset contains one representative motion type for each terrain skill, namely climbing onto a 50 cm box, vaulting over a 35 cm hurdle, jumping down from a 50 cm box, and stair ascent and descent on steps of approximately 20 cm height, together with omnidirectional walking motions such as straight walking and turning. To enlarge this limited initial dataset and improve terrain diversity, we further apply motion augmentation. In total, the resulting dataset contains approximately one hour of motion data for pre-training the motion generator and tracker.

1) Initial Dataset Construction: For motion skills recorded from videos, we first reconstruct human motions from raw videos using GVHMR [31]. We then retarget the reconstructed motions both from videos and motion datasets to the humanoid robot using a contact-constrained IK solver [32], similar to [12]. After obtaining the retargeted robot motions, we manually place terrain objects to align with the original motions and preserve the physical interaction between the robot and the environment. To ensure that the final training data are physically plausible for both the motion tracker and the motion generator, we do not directly use the raw retargeted trajectories; instead, we refine training motions from a DeepMimic-style tracking policy trained on those retargeted motions.

2) Motion Augmentation: To further improve the generalization of both the motion tracker and the motion generator, we enlarge the dataset through kinematics-based augmentation. Specifically, we augment existing motions by varying terrain geometry, including scaling obstacle heights and inserting random small boxes along the motion path, and then optimizing the motions to remain consistent with the modified terrain. The optimization includes losses such as terrain penetration and motion smoothness losses to reduce collisions and discontinuities [21]. As in the initial dataset construction, we do not use the optimized trajectories directly; instead, we train a tracking policy on them and record the resulting physically feasible trajectories. As a result, the augmented dataset covers box climbing and jump-down motions on 35–75 cm boxes, vaulting motions over 25–45 cm hurdles, stair motions on steps of 15–20 cm height, and omnidirectional walking motions with randomly placed small boxes.

B. Pre-training Stage

The pre-training stage equips both the motion generator and the motion tracker with the basic ability to react to different environments and to output appropriate motion references and joint actions.

1) Whole-body Motion Tracker: We first train a single whole-body motion tracker on a single RTX 4090 GPU using a DeepMimic-style RL framework with PPO in IsaacLab [33], based on the motion data collected in Sec. III-A. We design imitation rewards r_{mimic} and regularization terms r_{reg} (see Table I) so that the policy can imitate all reference motions while respecting the physical constraints of the real hardware:

$$R_{\text{pre}} = r_{\text{mimic}} + r_{\text{reg}}. \tag{1}$$

The tracker observation consists of the reference state, including linear and angular velocity in $\mathbb{R}^3$, joint position

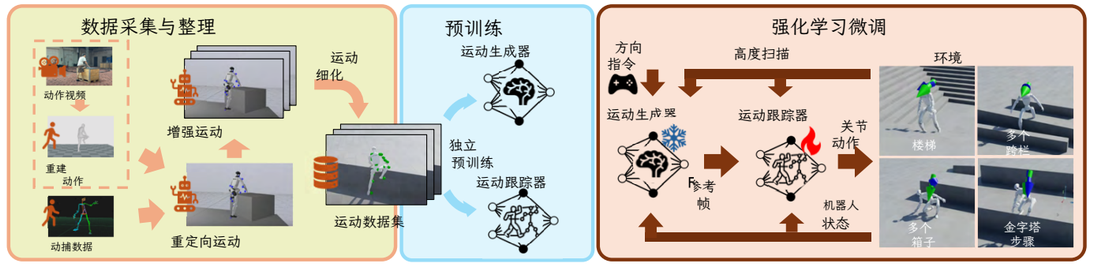


图2. 训练框架概览。(a) 数据采集与整理：通过运动重建与重定向，从人体运动获得全身机器人运动，随后进行运动增强。(b) 预训练：基于扩散的运动生成器和基于强化学习的运动跟踪器仅使用阶段(a)中的离线运动数据集进行训练。(c) 强化学习微调：运动跟踪器在随机化方向命令下，并在配置随机化的更多样化地形（包括楼梯、多个跨栏障碍物、箱体和金字塔形高台阶）上进行微调，同时运行冻结的运动生成器。

运动生成器，该生成器基于重定向的人类运动数据进行训练，作为技能组合模块，根据环境条件为低级跟踪策略生成相应的运动参考。此外，为了减少生成器引入的运动伪影的影响并提高控制性能，我们在随机化方向指令和更多样化的地形配置下，使用强化学习进一步微调运动跟踪策略，同时使用在离线运动数据集上预训练的冻结运动生成器。更多细节详见第III-C节。

III. 方法

整体框架如图2所示。我们的方法包含三个阶段：(a) 收集不同地形下的全身运动数据，并将其转换为动力学可行的机器人轨迹；(b) 在该数据集上预训练基于强化学习的运动跟踪器和基于扩散的运动生成器；(c) 在更多样化的地形配置上微调运动跟踪器，同时保持运动生成器冻结，并在部署时以滚动时域方式运行。我们将在下文对各阶段分别进行描述。

A. 数据收集与整理

我们的初始运动数据包含约5分钟的运动片段，来源有两个：我们自己拍摄的运动视频和公共运动数据集 [12, 30]。该数据集为每种地形技能包含一种代表性运动类型，即爬上50厘米高的箱子、跨越35厘米高的跨栏、从50厘米高的箱子上跳下，以及在约20厘米高的台阶上上下楼梯；此外还有直行、转弯等全向行走动作。为了扩充这一有限的初始数据集并提高地形多样性，我们进一步应用了运动增强。最终，得到的数据集包含约一小时的运动数据，用于预训练运动生成器和跟踪器。

1) 初始数据集构建：对于从视频中记录的运动技能，我们首先使用GVHMR [31]从原始视频中重建人体运动。然后，我们使用一个接触约束的IK求解器 [32]，（类似于 [12]）将来自视频和运动数据集的重建运动重定向到人形机器人上。在获得重定向后的机器人运动后，我们手动放置地形物体，以与原始运动对齐，并保留机器人与环境之间的物理交互。为确保最终训练数据对运动跟踪器和运动生成器都具有物理合理性，我们不直接使用原始的重定向轨迹；而是使用在这些重定向轨迹上训练的DeepMimic风格跟踪策略来细化训练运动。

2) 运动增强：为了进一步提升运动跟踪器和运动生成器的泛化能力，我们通过基于运动学的增强方法来扩充数据集。具体来说，我们通过改变地形几何来增强现有运动，包括缩放障碍物高度和在运动路径上随机插入小箱子，然后优化运动使其与修改后的地形保持一致。优化过程包括地形穿透损失和运动平滑损失等损失项，以减少碰撞和不连续性 [21]。与初始数据集构建一样，我们不直接使用优化后的轨迹，而是基于它们训练跟踪策略，并记录由此产生的物理可行轨迹。最终，增强后的数据集涵盖了在35–75厘米高的箱子上进行攀爬和跳下的动作、跨越25–45厘米高跨栏的动作、在15–20厘米高台阶上上下楼梯的动作，以及带有随机放置小箱子的全向行走动作。

B. 预训练阶段

预训练阶段使运动生成器和运动跟踪器都具备应对不同环境并输出合适运动参考与关节动作的基本能力。

1) 全身运动跟踪器：我们首先基于第 III-A 节收集的运动数据，在 IsaacLab [33], 中使用带 PPO 的 DeepMimic 风格强化学习框架，在单个 RTX 4090 GPU 上训练一个全身运动跟踪器。我们设计模仿奖励 r_{mimic} 和正则化项 r_{reg}（见表 I），使策略能够遵循真实硬件的物理约束，同时模仿所有参考运动：

$$R_{\text{pre}} = r_{\text{mimic}} + r_{\text{reg}}. \tag{1}$$

跟踪器的观测由参考状态组成，包括 $\mathbb{R}^3$中的线速度和角速度、关节位置

and velocity in $\mathbb{R}^{23}$, and key body positions in the base frame in $\mathbb{R}^{9\times3}$; proprioceptive terms, including five consecutive frames of base angular velocity, projected gravity in $\mathbb{R}^3$, joint position and velocity, and the previous action in $\mathbb{R}^{23}$; together with terrain height scans in $\mathbb{R}^{17\times11}$. The policy outputs 23-dimensional target joint positions as actions. Although Yang et al. [12] show that terrain information is not strictly required when the demonstrations are sufficiently accurate, we find that it can be beneficial to include exteroceptive input already at this stage, as it helps the later fine-tuning stage.

2) Diffusion-based Motion Generator: Our motion generator is based on a diffusion model that predicts whole-body reference motion sequences, following the architecture proposed in MDM [17, 21]. The model predicts future motion features over a 0.5-second horizon (25 frames), including root position in $\mathbb{R}^3$, root orientation in $\mathbb{R}^4$, joint positions in $\mathbb{R}^{23}$, and body link positions in $\mathbb{R}^{23\times3}$, conditioned on the target heading vector, terrain height scan, and the past two frames of motion features.

During training, we randomly sample motion sequences from the dataset, compute the heading vector from the base pose difference, and use the first two frames as conditions to predict the remaining frames. The reconstruction loss is then computed by comparing the predicted motion with the ground-truth sequence. In addition to the reconstruction loss, we incorporate geometric losses including velocity, joint consistency, and terrain penetration losses, similar to [21]. The generator is trained on the collected motion dataset and later serves as the motion planner that provides reference motions for the tracking policy over different terrains. To improve robustness against noisy robot states, we additionally perturb the height scans and previous-state conditions during training.

However, as discussed in Sec. II-B, generated motions may still contain artifacts, which can lead to failure when directly paired with a tracker trained mainly on high-quality references. Simply combining the pretrained motion generator and motion tracker can therefore result in limited robustness and reliability.

C. RL Fine-tuning Stage

In this stage, we improve the robustness and reliability of the pretrained motion generator and motion tracker, and adapt the full control system for real-time closed-loop deployment on a real humanoid robot. Directly combining the pretrained generator and tracker may still suffer from motion artifacts, tracking failures caused by imperfect references, and limited generalization to terrains and target directions beyond those covered by the training dataset. Therefore, we introduce an additional fine-tuning stage in which the motion tracker is fine-tuned with the frozen motion generator via RL on more diverse terrains and under varied target direction conditions.

During this stage, the motion generator produces reference frames conditioned on the past two frames of robot states to form a closed-loop motion prediction process, rather than relying on autoregressive feedback of its own previous predictions. To reduce deployment latency under limited onboard computation, we use only 2 denoising steps in the motion generator. We also inject additional noise into both the motion generator inference process and the tracker observations to improve robustness against real-world disturbances.

In addition, we randomize the target heading direction and introduce a heading tracking reward as a task reward r_{task} (see Table I), together with the imitation r_{mimic} and regularization terms r_{reg}, to encourage the full system to follow the desired direction while leveraging the motion priors acquired during pre-training:

$$R_{\text{post}} = r_{\text{mimic}} + r_{\text{reg}} + r_{\text{task}}. \tag{2}$$

We further introduce more diverse training terrains, including stairs with 15–25 cm step heights, continuous vaulting hurdles of 25–55 cm height, and multiple climbing boxes and pyramid steps ranging from 30–85 cm in height with varying widths as well as yaw and pitch angles.

After fine-tuning, the tracking policy effectively acts as a motion filter: it tracks the reference produced by the generator, while adjusting its execution to suppress unsafe behaviors using exteroceptive terrain observations. Despite the reduced number of denoising steps, which may further degrade raw generation quality, the fine-tuned tracker still robustly tracks the motion patterns produced by the generator and successfully traverses diverse terrains. Randomization of target headings and terrain configurations further enables behaviors beyond those present in the offline data, such as corner box climbing, reorientation on the box before descent, and continuous vaulting.

D. Details for Hardware Deployment

For hardware experiments, the entire pipeline is deployed fully onboard the Unitree G1 robot. We use DLIO [34] with the onboard Livox MID360 LiDAR and IMU to estimate the robot base pose, which is used as an input to motion generation. For terrain perception, we employ Elevation Mapping CuPy [35] together with the DLIO base pose to reconstruct terrain height information over different real-world environments.

Since the neck joint of the G1 robot is passive, aggressive motions can induce head movement that degrades base pose estimation. To improve state-estimation stability, we fuse the LiDAR IMU and torso IMU measurements by subtracting the orientation estimated by DLIO from the moving-average pitch measured by the torso IMU, which allows us to estimate the head pitch variation and compensate for its effect.

To accelerate diffusion-based motion generation, we deploy the generator with TensorRT, reducing inference time to approximately 0.02 s. Although the motion generator operates with a 0.5-second prediction horizon (2 Hz) during fine-tuning, we update the reference every 0.25 seconds in a receding-horizon way during deployment to improve responsiveness to the surrounding environment. The motion generator runs on a Jetson Thor mounted on the back of the robot, while the motion tracking controller and the remaining onboard modules run on the Jetson Orin integrated into the Unitree G1 platform.

IV. Experiments

In this section, we present the experimental results of the proposed humanoid locomotion system in simulation and on real hardware. We also provide quantitative evaluation to examine the contribution of motion generation to generalization, and the effect of RL fine-tuning of the motion tracker on robustness and success rates across terrain types.

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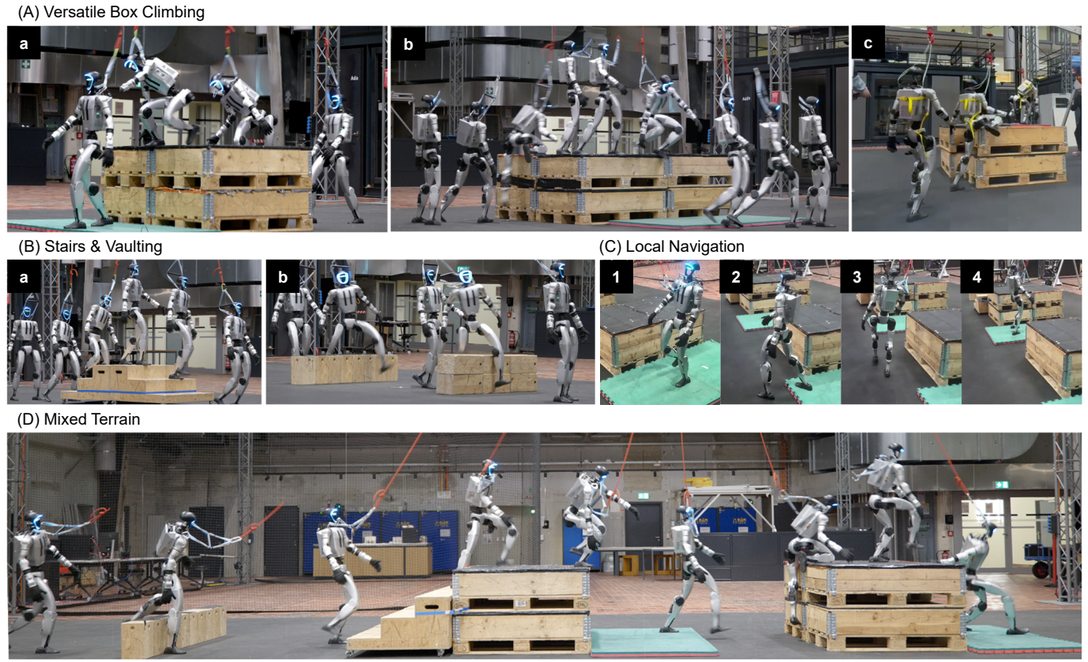


Fig. 3. Results of hardware experiments. (A) The robot climbs onto a 75 cm box and jumps down in three different ways: (a) straight ascent and descent; (b) straight ascent followed by a 90° right turn and side jump-down; and (c) ascent and descent from the box corner. (B) The robot traverses (a) a staircase and (b) a sequence of vaulting hurdles. (C) The robot shows local navigation behavior by bypassing the box to reach the target position. (D) The robot traverses a mixed terrain sequence consisting of a vaulting obstacle, stairs, and box climbing.

TABLE I
REWARD EQUATIONS FOR RL-BASED MOTION TRACKER

Group	Name	Equation
r_{mimic}	Base Pose Tracking	$3\exp\left(-4\left\|\mathbf{p}_b^w - \mathbf{p}_b^{w*}\right\|^2 - \left\|\mathbf{q}_b \otimes \mathbf{q}_b^{*-1}\right\|^2\right)$
	Base Velocity Tracking	$\exp\left(-\left\|\mathbf{v}_b^b - \mathbf{v}_b^{b*}\right\|^2 - 0.1\left\|\boldsymbol{\omega}_b^b - \boldsymbol{\omega}_b^{b*}\right\|^2\right)$
	Joint Position Tracking	$2.5\exp(-2\sum_{i=1}^{23} 0.25(q_i - q_i^*)^2)$
	Joint Velocity Tracking	$0.5\exp(-2\sum_{i=1}^{23} 0.01(\dot{q}_i - \dot{q}_i^*)^2)$
	Body Pos Tracking (Base)	$2\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}_{l,i}^b - \mathbf{p}_{l,i}^{b*}\right\|^2\right)$
	Body Pos Tracking (Global)	$\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}_{l,i}^w - \mathbf{p}_{l,i}^{w*}\right\|^2\right)$
r_{reg}	Action Rate (First-Order)	$-0.1\sum_{i=1}^{23}(a_{t,i} - a_{t-1,i})^2$
	Action Rate (Second-Order)	$-0.05\sum_{i=1}^{23}(a_{t,i} - 2a_{t-1,i} + a_{t-2,i})^2$
	Joint Position Limits	$-10\sum_{i=1}^{23}\max(q_i - q_{\text{lim},i}, 0)$
	Joint Velocity Limits	$-\sum_{i=1}^{23}\max(\dot{q}_i - \dot{q}_{\text{lim},i}, 0)$
	Torque Limits	$-0.001\sum_{i=1}^{23}\max(\tau_i - \tau_{\text{lim},i}, 0)$
	Joint Torques	$-0.00001\sum_{i=1}^{23}\tau_i^2$
	Body Linear Acceleration	$-0.01\sum_{i=1}^{9}\left\|\ddot{\mathbf{p}}_{l,i}^w\right\|^2$
r_{task}	Target Heading Tracking	$1.5\langle\mathbf{v}_b^b, \mathbf{d}_b\rangle / \left\|\mathbf{v}_b^b\right\|$

TABLE II
SYMBOLS USED IN TABLE I

Symbol	Description
$(\cdot)^*$	Reference quantity
$(\cdot)^b$	Expressed in the base frame
$(\cdot)^w$	Expressed in the world frame
$\mathbf{p}_b$	Base position
$\mathbf{q}_b$	Base orientation quaternion
$\mathbf{v}_b$	Base linear velocity
$\boldsymbol{\omega}_b$	Base angular velocity
$\mathbf{d}_b$	Base target heading
$\mathbf{p}_{l,i}$	Position of key body i
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A. Hardware Results

We evaluate the proposed pipeline on real hardware as shown in Fig. 1, 3, and the supplementary video. The hardware experiments demonstrate both the effectiveness of our controller and its generalization ability across diverse terrains. In particular, we evaluate the controller on a range of terrain types, including box ascent and descent, stair ascent and descent, and continuous vaulting over obstacles of varying heights, as well as on more challenging compound terrain settings that combine vaulting, stairs, and box climbing within a single run. For box ascent and descent, our controller exhibits versatile climbing behaviors, including box traversal from different approach directions as well as the ability to traverse a box with on-top reorientation before descent. During ascent, the robot uses its knees and hands for support, while during descent it jumps off the box and uses the hands for cushioning, consistent with the motion style observed in the training data. For vaulting, the robot can traverse multiple hurdles of different heights, typically crossing over them directly rather than stepping onto the hurdle first.

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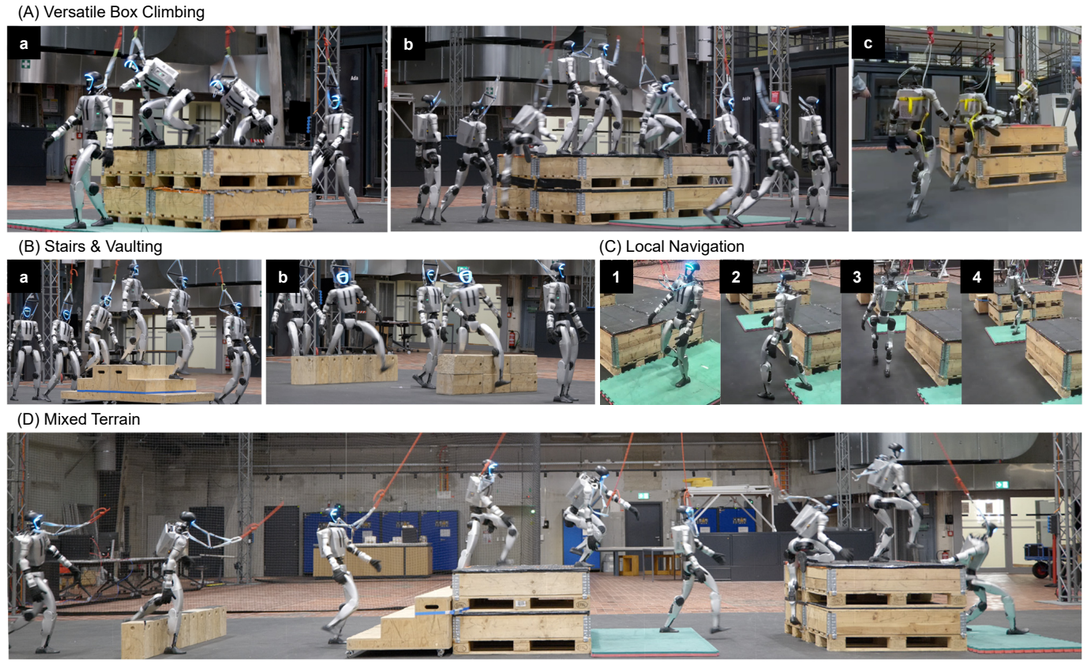


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TABLE III
COMPARATIVE SUCCESS RATES OF FIXED-REFERENCE MOTION TRACKING AND THE PROPOSED FULL SYSTEM ACROSS THREE EVALUATION SETTINGS.

(a) Box Climbing

Box Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
40	0	0.982	1.000
50	-30	0.988	0.980
	30	1.000	0.982
	-15	0.974	1.000
	15	0.970	0.998
	0	0.998	1.000
60	0	0.984	0.996
70	0	0.606	0.966
80	0	0.230	0.962
Mean ± SD		0.859 ± 0.252	**0.987 ± 0.014**

(b) Vaulting

Hurdle Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
25	0	0.978	1.000
35	-40	0.808	1.000
	40	0.492	1.000
	-20	0.922	1.000
	20	0.954	1.000
	0	0.988	1.000
45	0	0.950	1.000
55	0	0.348	0.920
Mean ± SD		0.805 ± 0.231	**0.990 ± 0.026**

(c) Ascending Stairs

Step Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
15	0	1.000	1.000
20	-30	0.948	1.000
	30	0.916	0.994
	-15	0.970	1.000
	15	0.990	0.996
	0	0.980	1.000
25	0	0.114	0.986
Mean ± SD		0.845 ± 0.300	**0.997 ± 0.005**

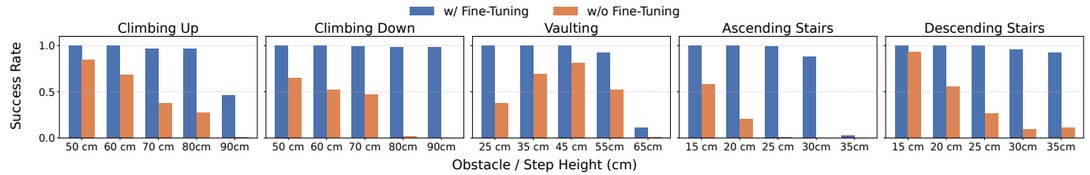


Fig. 4. Comparative results of the success rates of the motion tracker with and without fine-tuning across five terrain traversal tasks, including climbing up/down, vaulting, and stair ascent/descent. For each task, we spawn 500 robots in simulation with randomized initial poses and assign each of them a goal position that guides traversal of the corresponding terrain obstacle, from which the target direction is computed. The results consistently show that fine-tuning the tracker with a frozen motion generator improves the success rate of reaching the goal.

For stairs, the controller switches to stair ascent and descent behaviors. When these terrain types are combined, the controller can also transition dynamically among different motion styles according to the terrain, including sequences such as stair ascent following a jump-down motion, even though such terrain combinations are not seen during RL fine-tuning.

In addition, as shown in Fig. 3C and the supplementary video, we observe a form of local navigation behavior in the system after RL fine-tuning: when the robot is not in a suitable state to traverse an obstacle under the direction command computed from the given goal position using the motion generated by the planner, the tracking policy may partially override the reference motion and instead bypass the obstacle from the side to avoid failure and still reach the goal.

B. Quantitative Results

We further analyze two key design choices of our framework through quantitative simulation results: whether online motion generation improves adaptation and generalization across diverse terrains at deployment, and whether the additional fine-tuning stage is necessary after coupling the generator with the motion tracker. As shown in Table III and Fig. 4, both components are important for the final system.

1) Benefit of Online Motion Generation: We first evaluate whether online motion generation is beneficial for enabling the final system to adapt to diverse terrain variations by comparing it with fixed-reference tracking. We compare two settings on three representative tasks: box climbing, vaulting, and stair ascent. For box climbing, the reference trajectory is a motion for climbing onto a 50 cm box; for vaulting, the reference is a trajectory for traversing a 35 cm-high and 20 cm-wide hurdle; for stair ascent, the reference is a trajectory for ascending a staircase with 30 cm-wide and 20 cm-high steps. We then modify the terrain geometry at test time by changing the obstacle height or rotating the terrain mesh in yaw. **Tracker Only** uses the fine-tuned motion tracker with a fixed reference trajectory, and success is measured by whether the robot reaches the final XY base position of the reference trajectory. **Tracker + Gen.** uses the proposed full system, where the motion generator produces reference motions online from the target direction and terrain input, and success is measured by whether the robot reaches the goal position placed on the terrain. The goal position is chosen to match the final base position of the reference trajectory used in the *Tracker Only* setting, and the direction command is computed online accordingly. For each task, 500 robots are spawned in simulation with randomized initial poses and joint positions.

As shown in Table III, the full system with online motion generation achieves better generalization and higher average success rates across all three tasks than fixed-reference tracking on the corresponding terrain settings. For box climbing, the *Tracker Only* baseline performs well close to the nominal reference condition, but becomes brittle as the obstacle height increases. In contrast, *Tracker + Gen.* remains consistently robust across all tested settings. For vaulting, the fine-tuned tracker already exhibits some local generalization and can handle different hurdle heights, but its performance still degrades under larger changes in terrain height and orientation. The full system, by comparison, remains stable across all evaluated cases. A similar pattern appears in stair ascent: while the fixed-reference tracker can tolerate yaw perturbations, it deteriorates markedly under the height increase, whereas the full system continues to achieve nearly perfect performance.

These results highlight two points. First, the fine-tuned tracker is not entirely rigid. Because it observes terrain height scans and is fine-tuned on diverse terrain configurations, it can retain a limited degree of local terrain generalization. This is particularly visible in vaulting and box climbing, where nearby obstacle scales can sometimes be handled

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TABLE III
COMPARATIVE SUCCESS RATES OF FIXED-REFERENCE MOTION TRACKING AND THE PROPOSED FULL SYSTEM ACROSS THREE EVALUATION SETTINGS.

(a) Box Climbing

Box Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
40	0	0.982	1.000
50	-30	0.988	0.980
	30	1.000	0.982
	-15	0.974	1.000
	15	0.970	0.998
	0	0.998	1.000
60	0	0.984	0.996
70	0	0.606	0.966
80	0	0.230	0.962
Mean ± SD		0.859 ± 0.252	**0.987 ± 0.014**

(b) Vaulting

Hurdle Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
25	0	0.978	1.000
35	-40	0.808	1.000
	40	0.492	1.000
	-20	0.922	1.000
	20	0.954	1.000
	0	0.988	1.000
45	0	0.950	1.000
55	0	0.348	0.920
Mean ± SD		0.805 ± 0.231	**0.990 ± 0.026**

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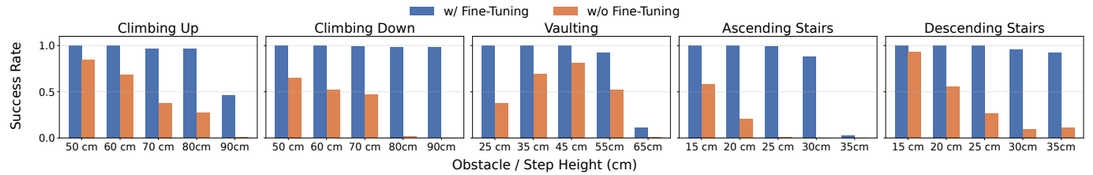


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even without adapting the reference motion. Second, this limited generalization is fundamentally insufficient for robust perceptive locomotion. Once the terrain geometry changes more substantially, especially in ways that require changing the timing or style of the motion rather than merely tolerating small mismatch, the fixed-reference tracker degrades rapidly. During deployment, terrain-aware reference motions produced online by the motion generator allow the full system to better adapt its traversal behavior to diverse obstacle geometries. Overall, the results in Table III highlight the importance of online motion generation for adaptation and generalization across diverse terrains in the final system.

2) Impact of Fine-Tuning the Motion Tracker: The above results show that online motion generation is important for generalization during deployment. We next ask whether the additional fine-tuning stage is also necessary, or whether the pretrained tracker can simply be combined with the motion generator directly. To answer this question, we compare the pretrained tracker and the fine-tuned tracker under the same motion generator, as shown in Fig. 4. We evaluate five terrain traversal tasks, namely climbing up, climbing down, vaulting, ascending stairs, and descending stairs. For each task, a goal position is placed behind the terrain obstacle, from which the direction command is computed, and 500 robots are spawned in simulation to measure the success rate.

Figure 4 shows that fine-tuning the tracker consistently improves success rates across all evaluated tasks, with the largest gains appearing on the more difficult terrain settings. For climbing up and climbing down, the improvement becomes more apparent as the box height increases, indicating that fine-tuning is especially important when the robot is required to react robustly to more demanding vertical transitions. For stair ascent and descent, the fine-tuned tracker also shows clear advantages on larger step heights. Vaulting is comparatively easier than the other tasks, but the fine-tuned tracker still remains more reliable overall. Taken together, the histogram suggests that fine-tuning the tracker with RL does not merely provide marginal refinement, but systematically improves robustness across a broad range of traversal tasks.

A likely reason is that the pretrained tracker is optimized on a fixed set of offline motion trajectories, which are relatively smooth and physically filtered. During deployment, however, the online references generated by the motion generator are conditioned on noisy robot state, target direction, and terrain observations, and may therefore contain temporal discontinuities or motion artifacts. This creates a distribution mismatch between the tracker training references and the actual reference stream encountered during control. Fine-tuning with the frozen generator is therefore crucial: it allows the tracker to adapt to the distribution and imperfections of generated references, learn how to make effective use of motion patterns for task completion, and suppress unsafe behaviors that could otherwise lead to failure. This observation is also consistent with the qualitative behavior discussed in Sec. III-C, where the fine-tuned tracker effectively acts as a motion filter over the generated references. Overall, Fig. 4 shows that the fine-tuning stage is an important part of making the combined generation-and-tracking framework robust in practice.

V. Conclusion

Overall, our results demonstrate that combining motion generation with motion tracking is an effective way to realize perceptive whole-body humanoid locomotion. The proposed system enables efficient traversal over diverse terrain types and terrain combinations unseen during training, while the quantitative analysis highlights the importance of both online motion generation and tracker fine-tuning for adaptation and robustness.

One limitation of the current system is that it relies on a LiDAR-based elevation mapping pipeline to reconstruct the surrounding environment. When the mapping quality degrades due to sensing noises, the locomotion performance can deteriorate substantially. A promising direction for future work is to adopt perception architectures that do not depend as strongly on high-quality geometric sensing, such as neural mapping [6], or to incorporate a belief encoder [2] that compensates for sensing failures through proprioceptive observations. Another important direction is to extend the framework beyond pure locomotion to a broader range of loco-manipulation tasks and to more challenging outdoor environments.

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